

UNIVERSITY OF ST ANDREWS

CS4302 PRACTICAL 3

Image Processing

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Goal

The goal of this practical was to learn and apply image processing techniques using Matlab[®].

Tasks

The scripts are separated by task given in the practical specification. Where useful, scripts were created to hold functions that were used repeatedly. To run each practical task in order, the script *practical.m* can be used (This will generate a large number of figures). The images for processing are shown in figures 1, 2, and 3.



Figure 1: Image with large variation in colour and spatial frequency.

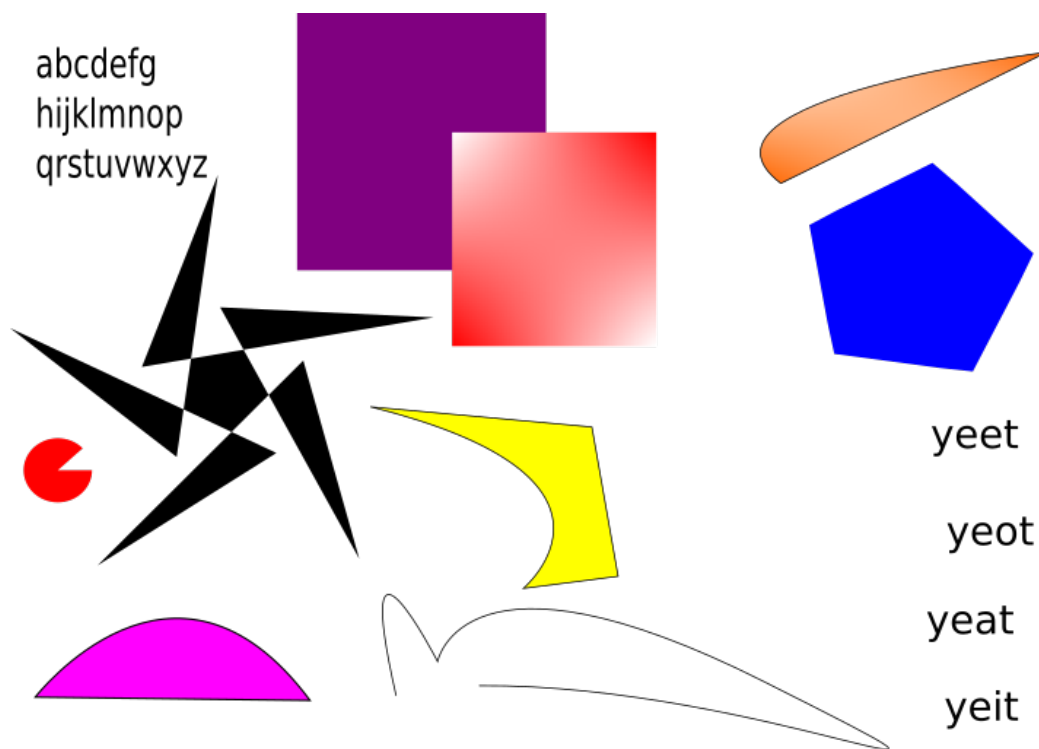


Figure 2: Vector image featuring text, polygons, and bezier curves. Originally created in Inkscape and exported as a PNG file.



Figure 3: Simple image with little variation in colour or spatial frequency.

1 RGB

For task 1, the function *showRGB.m* was created. After reading the files using *imread*, the three colour planes were separated. These were then all plotted individually for each image file. Since each colour has 256 values (given 8 bits each), the colourbar was supplied the range 0 to 256.

Figure 1 was filtered into figures 4, 5, and 6. Lanterns in the image that are most saturated are replaced with black in all other colour filtered images. For example, the blue lanterns roughly in the centre of the blue channel image are not visible in the red channel image. Since white is composed of all three colours, the white circles in the centre of the blue lanterns is still visible in each image. A similar effect can be seen in each of the other images.

Figure 3 was filtered into figures 7, 8, and 9. Since the image is mostly white, there is very little difference in the three channels.

Figure 12 was filtered into figures 10, 11, and 12. The rasterized vector image has the most obvious differences compared to the original image. The values of 0 are used for all three channels for the originally transparent background, and so the black text is lost. In the red image (Figure 10), the blue pentagon is lost. The varying shades of pink in the image appear to be the result of increasing the brightness (the 8-bit value) of the red and blue channels, especially the red. The square with the colour gradient applied appears a solid colour in the red and blue channels, showing that the green and blue channels are used to produce the white corners.

2 YCbCr

For this task, I used both inbuilt *rgb2ycbcr* and *ycbcr2rgb* methods, and then implemented my own respective versions *rgbToYCbCr* and *YCbCrToRGB* in order to better understand the transformation. The equations for the translation between the two colour spaces are shown in equations 1 and ??.

$$x = y + 1 \tag{1}$$

$$x = y + 2 \tag{2}$$

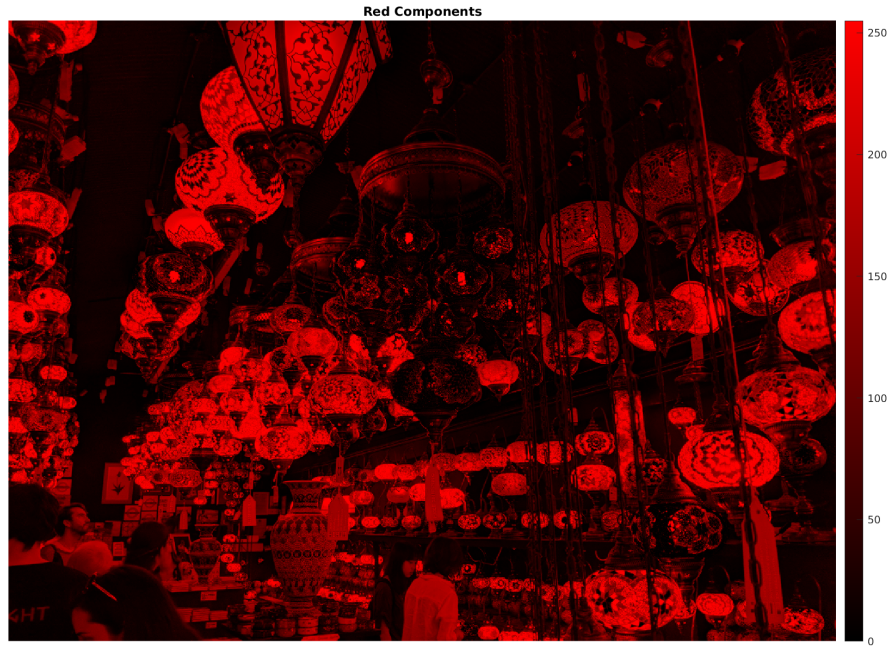


Figure 4: Red component of Figure 1.

Conclusion

This practical was incredibly enjoyable (excluding the whine), and greatly helped my understanding of Matlab. Actually implementing the technologies discussed in lectures resulted in many rabbit holes of learning, and a better understanding of the content. For example, the various types of audio filters and their pros and cons, and why it isn't as simple as 'blocking' the unwanted frequencies.

I realised only close to the deadline that since on the schools Windows installs, the older version of Matlab (2017a) is used, and it did not have the bandpass function yet. However, in 2018b, it does exist, and so the scripts will only work on the linux machines. Given more time, I would have liked to replace the bandpass function with the more specific filters available, such as elliptical and chebyshev filters.

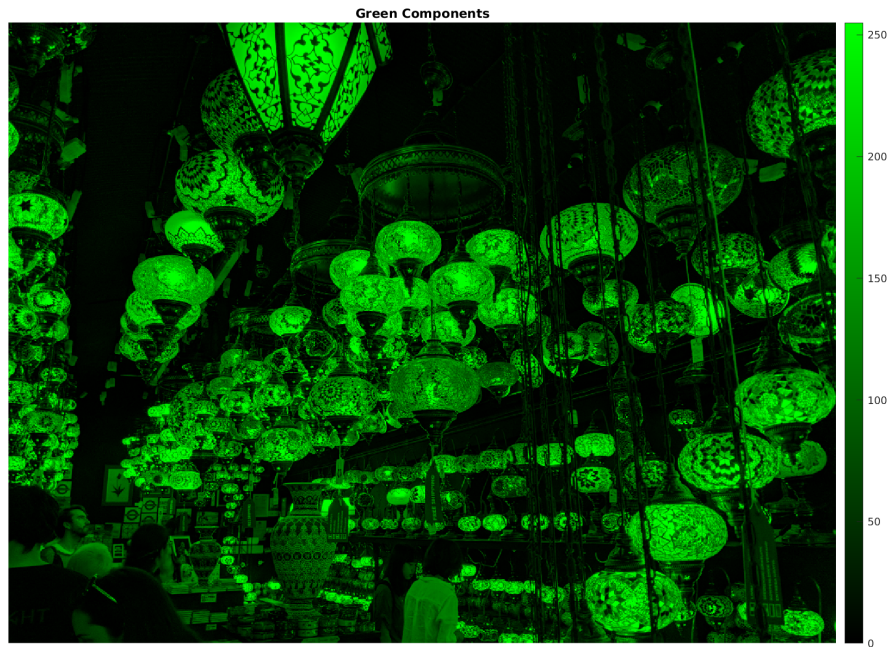


Figure 5: Green component of Figure 1.

References

- [1] Matlab documentation. <https://uk.mathworks.com/help/matlab/ref/audioplayer.html>. Accessed: 2018-11-2.
- [2] Kazumi Totaka. Mii channel music, 2006.
- [3] Frequencies for equal-tempered scale. <https://pages.mtu.edu/~suits/notefreqs.html>. Accessed: 2018-11-2.
- [4] Stephen Butterworth. On the theory of filter amplifiers. *Wireless Engineer*, 7:536–541, 1930.

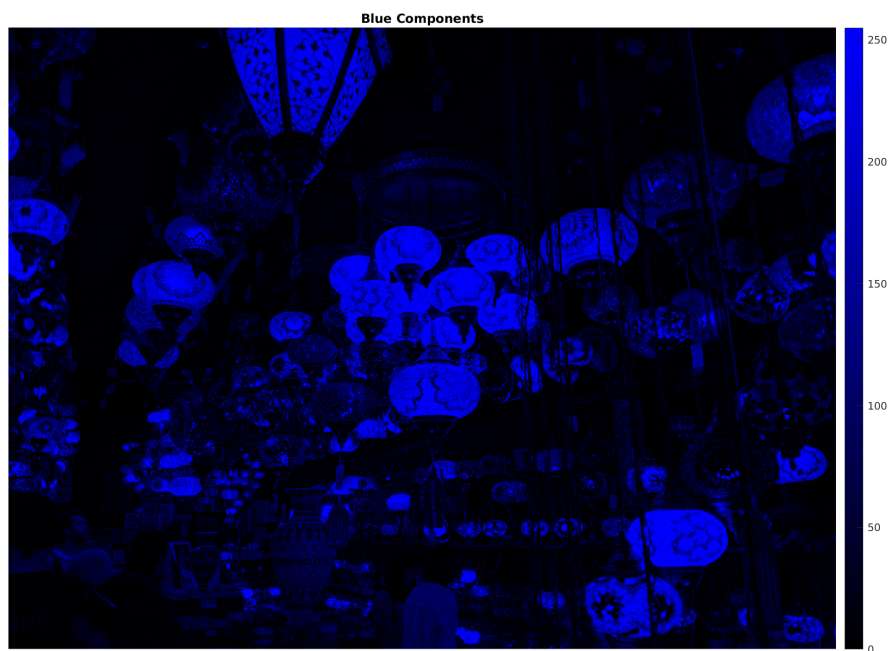


Figure 6: Blue component of Figure 1.

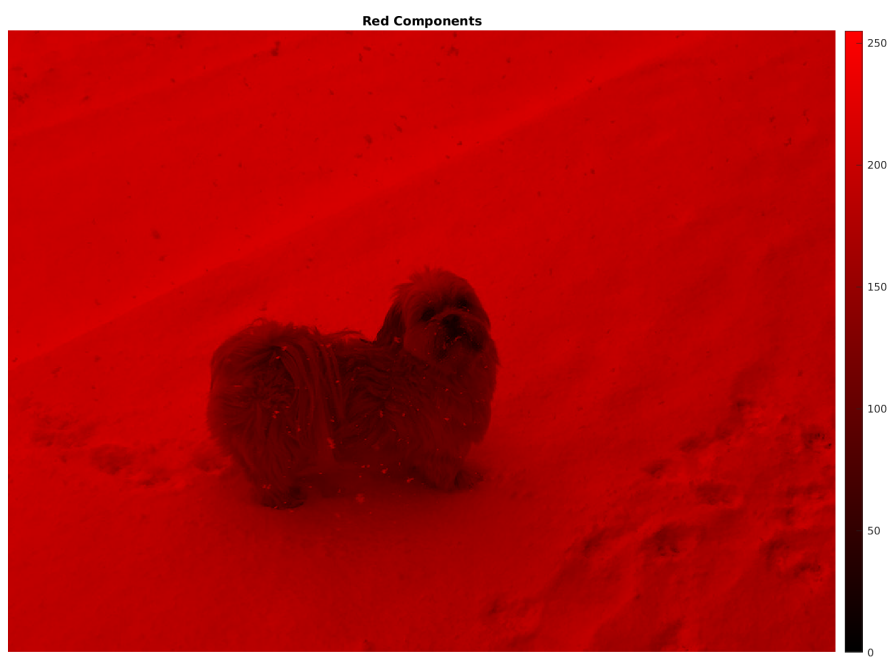


Figure 7: Red component of Figure 3.



Figure 8: Green component of Figure 3.

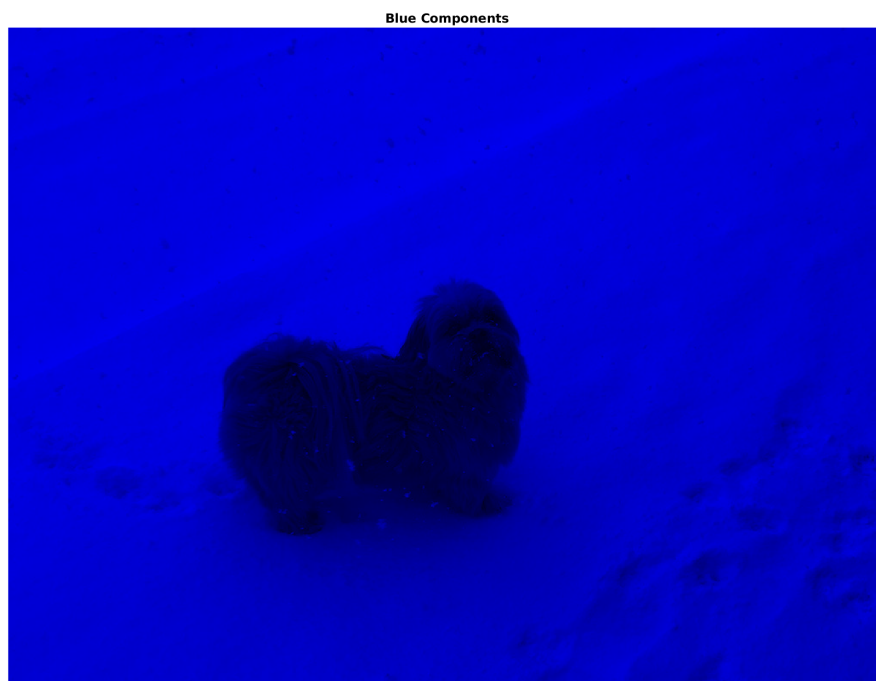


Figure 9: Blue component of Figure 3.

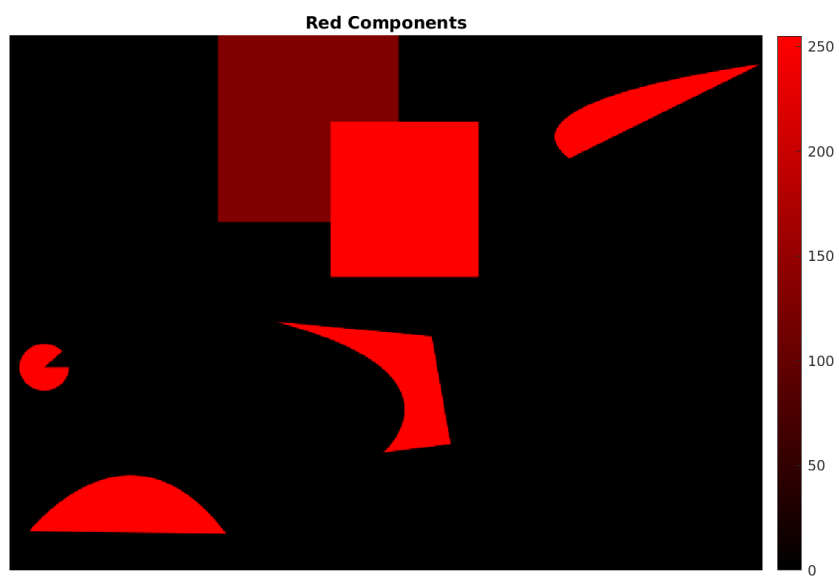


Figure 10: Red component of Figure 2.

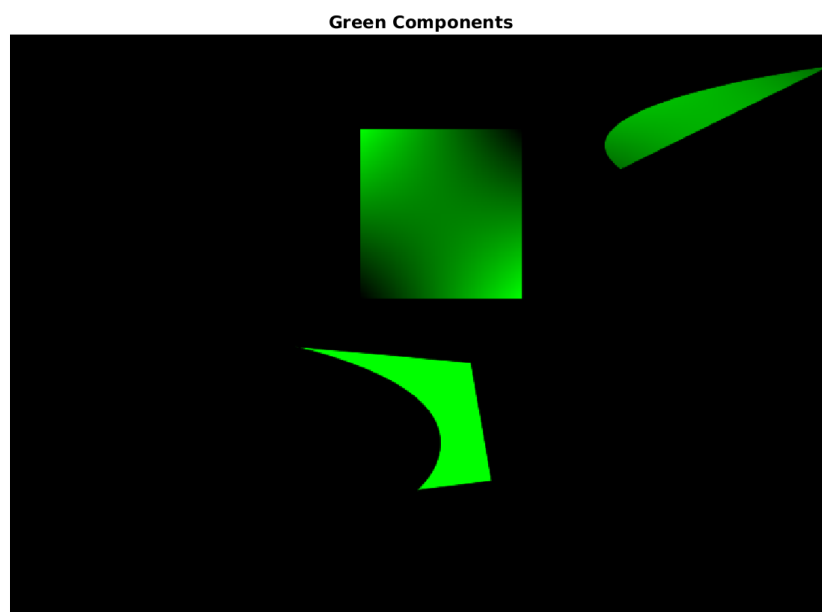


Figure 11: Green component of Figure 2.

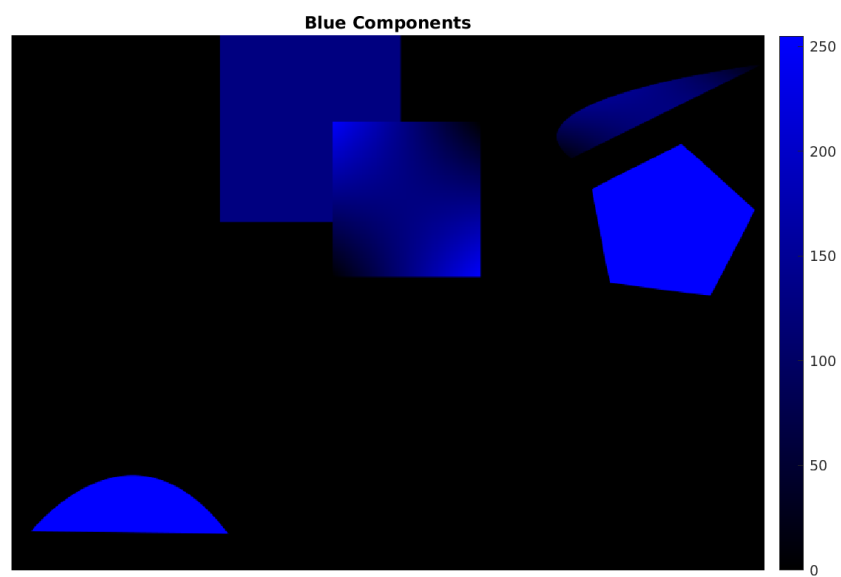


Figure 12: Blue component of Figure 2.